

Prilocaine: Drug information - UpToDate

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For additional information see "Prilocaine: Patient drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- 4% Citanest Plain Dental

Brand Names: Canada

- 4% Citanest Plain Dental

Pharmacologic Category

- Local Anesthetic

Dosing: Adult

Dental anesthesia

Dental anesthesia: Infiltration or conduction block: Initial: 40 to 80 mg (1 to 2 mL) as a 4% solution. AAPD guidelines, 2009 maximum recommended dose within a 2-hour period:

<70 kg: 6 mg/kg (400 mg)

≥70 kg: 400 mg or 5 to 6 cartridges

Note: The effective anesthetic dose varies with procedure, intensity of anesthesia needed, duration of anesthesia required and physical condition of the patient. Always use the lowest effective dose along with careful aspiration.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely due to limited duration of use.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. Undergoes hepatic metabolism; use with caution.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

Dental anesthesia

Dental anesthesia: Infiltration or conduction block:

Children <10 years: Doses >40 mg (1 mL) as a 4% solution per procedure rarely needed for procedures involving a single tooth, in a maxillary infiltration for 2 to 3 teeth, or for an entire quadrant with a mandibular block.

Children ≥10 years and Adolescents: Refer to adult dosing.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Undergoes renal metabolism; use with caution.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Undergoes hepatic metabolism; use with caution.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Degree of adverse effects in the central nervous system and cardiovascular system are directly related to the blood levels of local anesthetic. The effects below are more likely to occur after systemic administration rather than infiltration.

Frequency not defined:

Cardiovascular: Bradycardia, cardiac arrest, cardiovascular signs and symptoms (stimulation/depression), circulatory shock, edema, hypotension

Central nervous system: Apprehension, confusion, convulsions, dizziness, drowsiness, euphoria, localized warm feeling, loss of consciousness, nervousness, numbness, oral paresthesia (may be persistent), sensation of cold, twitching

Dermatologic: Dermal ulcer, urticaria

Gastrointestinal: Vomiting

Hematologic & oncologic: Methemoglobinemia

Hypersensitivity: Anaphylactoid reaction, hypersensitivity reaction

Neuromuscular & skeletal: Tremor

Ophthalmic: Blurred vision, diplopia

Otic: Tinnitus

Respiratory: Respiratory arrest, respiratory depression

Contraindications

Hypersensitivity to local anesthetics of the amide type or any component of the formulation; patients with congenital or idiopathic methemoglobinemia

Canadian labeling: Additional contraindications (not in US labeling): Citanest Plain Dental: Severe shock or heart block; inflammation or sepsis in the region of proposed injection

Warnings/Precautions

Concerns related to adverse effects:

- **CNS toxicity:** Careful and constant monitoring of the patient's state of consciousness should be done following each local anesthetic injection; at such times, restlessness, anxiety, tinnitus, dizziness, blurred vision, tremors, depression, or drowsiness may be early warning signs of CNS toxicity. Treatment is primarily symptomatic and supportive.
- **Methemoglobinemia:** Has been reported with local anesthetics; clinically significant methemoglobinemia requires immediate treatment along with discontinuation of the anesthetic and other oxidizing agents. Onset may be immediate or delayed (hours) after anesthetic exposure. Patients with glucose-6-phosphate dehydrogenase deficiency, congenital or idiopathic methemoglobinemia, cardiac or pulmonary compromise, exposure to oxidizing agents or their metabolites, or infants <6 months of age are more susceptible and should be closely monitored for signs and symptoms of methemoglobinemia (eg, cyanosis, headache, rapid pulse, shortness of breath, lightheadedness, fatigue). Use is contraindicated in patients with congenital or idiopathic methemoglobinemia.
- **Respiratory arrest:** Local anesthetics have been associated with rare occurrences of sudden respiratory arrest.
- **Seizures:** Convulsions due to systemic toxicity leading to cardiac arrest have also been reported, presumably following unintentional intravascular injection.

Disease-related concerns:

- **Cardiovascular disease:** Use with caution in patients with cardiovascular disease, severe shock, or heart block.
- **Familial malignant hyperthermia:** Prilocaine may potentially trigger malignant hyperthermia; follow standard protocol for identification and treatment.
- **Hepatic impairment:** Use with caution in patients with hepatic impairment; amide-type anesthetics are metabolized hepatically.

Special populations:

- **Acutely ill patients:** Use with caution in acutely-ill patients; reduce dose consistent with age and physical status.
- **Debilitated patients:** Use with caution in debilitated patients; reduce dose consistent with age and physical status.
- **Older adult:** Use with caution in the elderly; reduce dose consistent with age and physical status.
- **Pediatric:** Use with caution in children; reduce dose consistent with age and physical status.

Other warnings/precautions:

- **Administration:** Intravascular injections should be avoided; aspiration should be performed prior to administration; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided.
- **Appropriate dosing:** To avoid serious adverse effects and high plasma concentrations, the lowest dosage resulting in effective anesthesia should be administered. Repeated doses may significantly increase blood concentrations of both the drug or its metabolites; tolerance to elevated blood concentrations varies with patient status. Reduced dosages, commensurate with age and physical condition, should be given to patients who are debilitated, elderly, acutely-ill, or pediatric.
- **Trained personnel:** Dental practitioners using local anesthetic agents should be well trained in diagnosis and management of emergencies that may arise from the use of these agents. Resuscitative equipment, oxygen, and other resuscitative drugs should be available for immediate use.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

4% Citanest Plain Dental: 4% (1.8 mL)

Generic Equivalent Available: US

May be product dependent

Pricing: US

Solution (Citanest Plain Dental Injection)

4% (per mL): \$0.53

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

4% Citanest Plain Dental: 4% (1.8 mL)

Storage/Stability

Store at 25°C (77°F). Do not freeze.

Use: Labeled Indications

Local anesthesia: Production of local anesthesia in dentistry by nerve block or infiltration techniques.

Medication Safety Issues

Sound-alike/look-alike issues:

Prilocaine may be confused with Polocaine, PriLOSEC

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Articaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

BUPIvacaine (Liposomal): Local Anesthetics may increase adverse/toxic effects of BUPivacaine (Liposomal). Management: Liposomal bupivacaine should not be administered with local anesthetics, but may be administered 20 minutes or more after lidocaine. Avoid all local anesthetics within 96 hours after administration of liposomal bupivacaine. *Risk X: Avoid*

BUPIvacaine: Local Anesthetics may increase adverse/toxic effects of BUPivacaine. Management: Avoid using any additional local anesthetics within 96 hours after insertion of the bupivacaine implant (Xaracoll) or bupivacaine and meloxicam periarticular solution (Zynrelef) or within 168 hours after subacromial infiltration (Posimir brand). *Risk C: Monitor*

Dapsone (Topical): May increase adverse/toxic effects of Methemoglobinemia Associated Agents. *Risk C: Monitor*

Hyaluronidase: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Local Anesthetics: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Prilocaine. Combinations of these agents may increase the likelihood of significant methemoglobinemia. Management: Monitor for signs of methemoglobinemia when prilocaine is used in combination with other agents associated with development of methemoglobinemia. Avoid use of these agents with prilocaine/lidocaine cream in infants less than 12 months of age. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Local Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nitric Oxide: May increase adverse/toxic effects of Methemoglobinemia Associated Agents. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Primaquine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Primaquine. Specifically, the risk for methemoglobinemia may be increased. Management: Avoid concomitant use of primaquine and other drugs that are associated with methemoglobinemia when possible. If combined, monitor methemoglobin levels closely. *Risk D: Consider Therapy Modification*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Local Anesthetics. *Risk C: Monitor*

ROPivacaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Technetium Tc 99m Tilmanocept: Coadministration of Local Anesthetics and Technetium Tc 99m Tilmanocept may alter diagnostic results. Management: Avoid mixing and simultaneously co-injecting technetium Tc 99m tilmanocept with local anesthetics. This interaction does not appear to apply to other uses of these agents in combination. *Risk C: Monitor*

Pregnancy Considerations

Adverse events have not been observed in animal reproduction studies. Prilocaine crosses the placenta.

Breastfeeding Considerations

It is not known if prilocaine is excreted in breast milk. The manufacturer recommends that caution be exercised when administering prilocaine to nursing women.

Mechanism of Action

Local anesthetics bind selectively to the intracellular surface of sodium channels to block influx of sodium into the axon. As a result, depolarization necessary for action potential propagation and subsequent nerve function is prevented. The block at the sodium channel is reversible. When drug diffuses away from the axon, sodium channel function is restored and nerve propagation returns.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Infiltration: <2 minutes; Inferior alveolar nerve block: <3 minutes

Duration: Infiltration: Complete anesthesia for procedures lasting 20 minutes; Inferior alveolar nerve block: ~2.5 hours

Distribution: 190 to 260 L; crosses blood-brain barrier

Protein binding: 40% to 55% (α_1 acid glycoprotein)

Metabolism: Primarily metabolized hepatically and to a lesser extent renally; prilocaine hydrolyzed by amidases to produce ortho-toluidine and N-propylalanine; these compounds may also undergo ring hydroxylation

Half-life elimination: 1.6 hours; may be prolonged with hepatic or renal impairment

Excretion: Urine (<5% as unchanged drug)

Patient Counseling Points

(For additional information see "Prilocaine: Patient drug information")

What is this drug used for?

- It is used before dental care to numb the area.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Methemoglobinemia like a blue or gray color of the lips, nails, or skin; a heartbeat that does not feel normal; seizures; very bad dizziness or passing out; very bad headache; feeling very sleepy; feeling tired or weak; or shortness of breath
- Too much acid in the blood (acidosis) like confusion; fast breathing; fast heartbeat; a heartbeat that does not feel normal; very bad stomach pain, upset stomach, or throwing up; feeling very sleepy; shortness of breath; or feeling very tired or weak
- Numbness or tingling in or around the mouth that lasts or gets worse
- Dizziness or passing out
- Feeling lightheaded, sleepy, confused, or having blurred eyesight
- Slow heartbeat
- A heartbeat that does not feel normal
- Sweating a lot
- Ringing in ears
- Low mood (depression)
- Very nervous and excitable
- Trouble breathing, slow breathing, or shallow breathing
- Seizures
- Shakiness
- Twitching
- Change in balance
- Restlessness
- Anxiety
- Change in speech
- Feeling hot or cold

- Metallic taste
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AU) Australia: Citanest;
- (CO) Colombia: Citanest | Pricanest;
- (EC) Ecuador: Pricanest;
- (FI) Finland: Citanest;
- (GB) United Kingdom: Citanest;
- (IE) Ireland: Citanest;
- (MY) Malaysia: Citanest;
- (NL) Netherlands: Citanest;
- (NO) Norway: Citanest;
- (PE) Peru: Pricanest

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